

Spike-Timing-Based Classification: A Step-by-Step Computational Study of Tempotron Capacity

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Abstract

We build and validate a simulation pipeline for the spike-timing classifier capacity of Rubin, Monasson & Sompolinsky (RMS, 2010), in which the tempotron storage capacity α_c grows with the temporal-resolution parameter $K = T/\sqrt{\tau_s \tau_m}$ as $\alpha_c \simeq (\ln \ln K)/(2 \ln 2)$, exceeding the perceptron's $\alpha_c = 2$. We define the model, develop each pipeline component — the postsynaptic-potential kernel, the random spike-pattern ensemble, the decision rule and its threshold calibration, and the learning rule — with the theory each rests on, and verify every component against its closed form. We then calibrate the full measurement procedure against problems with known answers: it recovers the exact perceptron capacity $\alpha_c = 2$ and the Gardner margin across input dimensions $N \leq 500$ using two independent separability oracles, and reproduces the Gumbel extreme-value statistics of the peak voltage and the kernel closed form to numerical precision. A study of the threshold-weight-magnitude relationship probes the distinction between the *existence* of zero-error solutions and their *findability* by a gradient rule on the shattered tempotron solution space — the question that motivates the program.

Contents

1	Introduction	2
2	The tempotron model	2
3	The postsynaptic-potential kernel and forward pass	3
4	The random spike-pattern ensemble	4
5	Decision rule and threshold calibration	4
5.1	Convexity and the shattered solution space	5
5.2	Extreme-value theory and the threshold scale	6
5.3	Internal representations and the capacity	6
6	The learning rule	6
6.1	The rule is gradient descent	7
6.2	A subgradient on a piecewise-smooth surface	7
6.3	Existence versus findability	7
6.4	Momentum and the effective step	7

7	Validation of the capacity-measurement methodology	8
7.1	Perceptron capacity	8
7.2	Extreme-value statistics of the decision variable	10
7.3	The postsynaptic kernel	12
7.4	Summary	13
8	Tempotron memory capacity and the finite-size masking of the temporal lift	13
9	Findability, solution geometry, and noise robustness	15
10	The threshold–weight-magnitude relationship	16
11	Discussion and open directions	18
12	Reproducibility and code map	19

1 Introduction

The tempotron [1] is a leaky integrate-and-fire neuron used as a binary classifier of *spatio-temporal* spike patterns. Unlike the perceptron, whose decision is a single linear threshold on a static input vector, the tempotron’s decision is a threshold on the *maximum over time* of a temporally-extended membrane voltage. Rubin, Monasson & Sompolinsky (RMS) [2] showed, via statistical mechanics and extreme-value theory, that this maximum-over-time lifts the storage capacity above the perceptron’s and fractures the solution space into many small clusters.

The question motivating this study concerns the *learning surface*: how and why does a gradient-based learning rule reliably reach a zero-error weight vector when the feasible set is shattered into many disconnected clusters, and is gradient descent the right tool? This separates *existence* — the Gardner volume of solutions, which capacity theory computes — from *findability*, whether the learning dynamics land in it. Probing this empirically requires a simulation pipeline whose every component is independently verified, because a capacity is *inferred* through a measurement procedure that can be biased in ways invisible when the answer is unknown.

This paper builds and validates that pipeline. Section 2 defines the model; Section 3–6 develop and verify each component (kernel, stimulus ensemble, decision rule, learning rule) together with the theory each rests on; Section 7 calibrates the full procedure against problems with known answers (the perceptron capacity, the extreme-value statistics of the decision variable, and the kernel closed form); Section 10 reports a study of the threshold–weight-magnitude relationship; and Section 11 returns to the findability question. The capacity sweep $\alpha_c(K)$ itself is the subject of ongoing work and is not claimed here.

2 The tempotron model

A tempotron has N afferents with weights w_i . Afferent i fires spikes at times $\{t_i^f\}$. The subthreshold membrane voltage is

$$V(t) = V_{\text{rest}} + \sum_{i=1}^N w_i \sum_{t_i^f < t} K(t - t_i^f), \quad (1)$$

with the normalised postsynaptic potential (PSP) kernel introduced in Section 3. We take $V_{\text{rest}} = 0$ and measure voltage in units of the firing threshold. It is convenient to write $V(t; w) = w \cdot s(t)$

with the per-afferent PSP waveform

$$s_i(t) = \sum_{t_i^f} K(t - t_i^f) \geq 0, \quad (2)$$

which is *fixed* once a pattern's spikes are drawn.

Symbol	Meaning	Units
N	number of afferents (synapses)	—
w_i	synaptic weight	threshold units
τ_m	membrane time constant	ms
τ_s	synaptic time constant ($\tau_m > \tau_s > 0$)	ms
T	pattern window	ms
$K = T/\sqrt{\tau_s \tau_m}$	temporal resolution	—
$\alpha = P/N$	load (P patterns)	—
y	label	$\{-1, +1\}$
U_{th}	firing threshold	threshold units

Remark. The PSP *kernel* is $K(t)$ in the original papers, but RMS also write the *capacity* parameter as $K = T/\sqrt{\tau_s \tau_m}$. We keep the kernel callables named `psp_kernel/peak_time` in code and reserve the symbol K for resolution here.

3 The postsynaptic-potential kernel and forward pass

The normalised double-exponential PSP kernel is

$$K(t) = V_0 \left(e^{-t/\tau_m} - e^{-t/\tau_s} \right), \quad t \geq 0; \quad K(t) = 0, \quad t < 0, \quad (3)$$

with $\tau_m > \tau_s$. It is causal ($K(0) = 0$), rises, then decays. The prefactor V_0 fixes the peak to 1, so one spike on a unit-weight synapse contributes a peak EPSP of exactly the threshold.

Peak time. Setting $K'(t) = 0$,

$$\frac{1}{\tau_s} e^{-t/\tau_s} = \frac{1}{\tau_m} e^{-t/\tau_m} \implies \boxed{t_{\text{max}}^{(1)} \equiv t_{\text{peak}} = \frac{\tau_m \tau_s}{\tau_m - \tau_s} \ln \frac{\tau_m}{\tau_s}} \quad (4)$$

which is a maximum since $K''(t_{\text{peak}}) < 0$.

Normalisation. Requiring $K(t_{\text{peak}}) = 1$,

$$V_0 = \left[e^{-t_{\text{peak}}/\tau_m} - e^{-t_{\text{peak}}/\tau_s} \right]^{-1}. \quad (5)$$

For the standard ratio $\tau_m = 4\tau_s$ ($\tau_m = 15$ ms, $\tau_s = 3.75$ ms): $t_{\text{peak}} = 5 \ln 4 \approx 6.9315$ ms and $V_0 \approx 2.1165$.

Verification. The closed forms are confirmed two ways. Symbolically, solving $K'(t) = 0$ returns (4) exactly with $K''(t_{\text{peak}}) < 0$. Numerically, on a $1 \mu\text{s}$ grid the kernel attains its global maximum of 1 at t_{peak} to within 10^{-12} , is causal, and the forward pass (1) for one unit spike peaks at 1 at $t_f + t_{\text{peak}}$. (Section 7.3 extends this to a grid of (τ_m, τ_s) .)

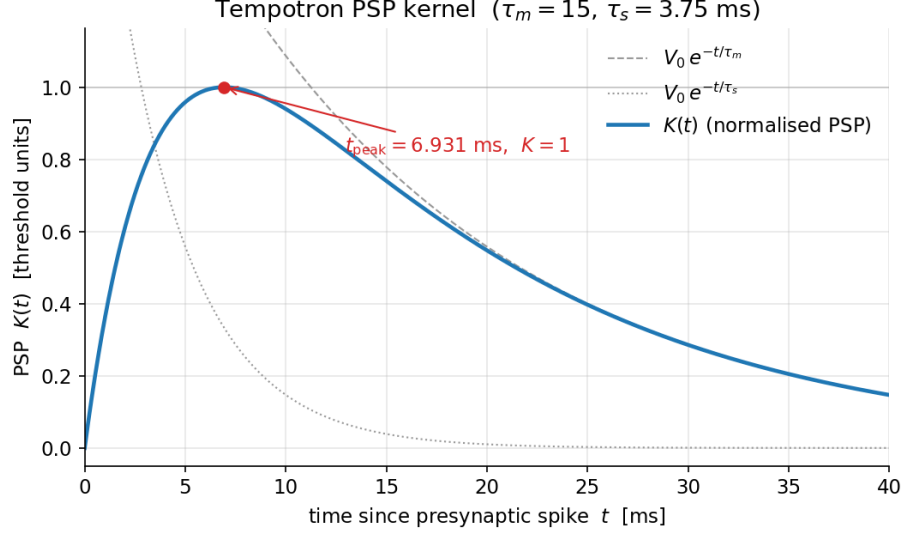


Figure 1: Normalised PSP kernel (3), peaking at 1 at t_{peak} (4), shown with its two exponential components.

4 The random spike-pattern ensemble

RMS classify a batch of $P = \alpha N$ patterns. In each pattern every afferent emits a **homogeneous Poisson spike train on $[0, T]$ with rate $1/T$** , and the label $y \in \{-1, +1\}$ is i.i.d. with equal probability [2].

Consequences of rate $1/T$. The mean count per afferent is $\lambda T = 1$, so the count is $\text{Poisson}(1)$ *independently of T* : $\approx 37\%$ of afferents are silent, some fire twice or more. Increasing T does not change the count statistics; it spreads the same ~ 1 spike over a wider window. This is how the resolution enters,

$$K = \frac{T}{\sqrt{\tau_s \tau_m}}, \quad (6)$$

counting the distinguishable temporal slots within the PSP width $\sqrt{\tau_s \tau_m}$; K is swept via T at fixed kernel timescales (with $\tau_s = \tau_m/4$).

Verification. For a homogeneous Poisson process on $[0, T]$ the count is $n \sim \text{Poisson}(1)$ with $\mathbb{E}[n] = \text{Var}[n] = 1$, the times given the count are i.i.d. $\mathcal{U}[0, T]$, and labels are balanced. The generated ensemble reproduces these: measured mean and variance of the per-afferent count are 1.0018 and 1.0013 (Fig. 2), the Kolmogorov–Smirnov distance of the rescaled spike times to $\mathcal{U}[0, 1]$ is below 0.02, and the labels are balanced.

5 Decision rule and threshold calibration

Ignoring post-spike reset [2], the binary decision is whether V crosses threshold anywhere in $[0, T]$, i.e.

$$\text{output} = \text{sign}(V_{\text{max}} - U_{\text{th}}), \quad V_{\text{max}} = \max_{t \in [0, T]} V(t), \quad t_{\text{max}} = \arg \max_t V(t). \quad (7)$$

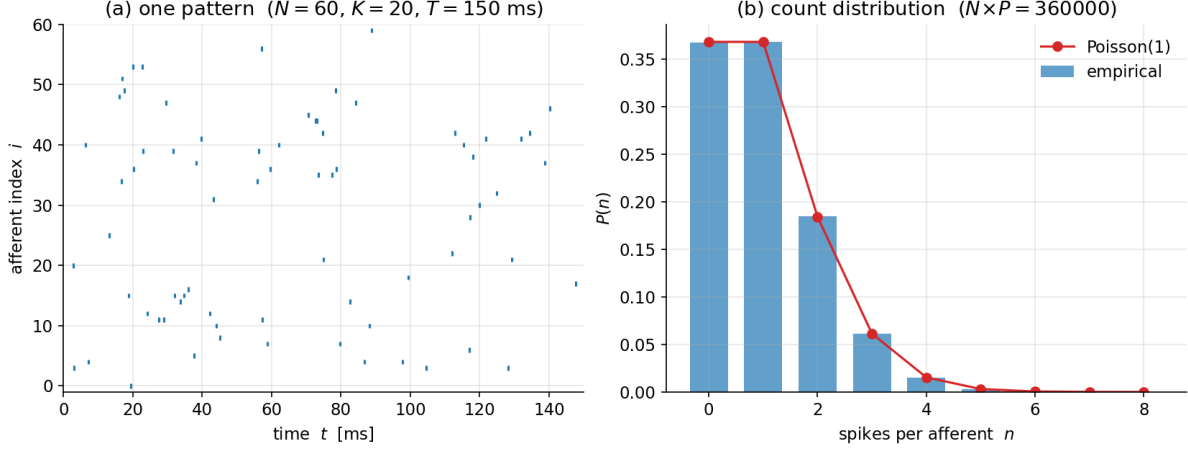


Figure 2: (a) Raster of one pattern: sparse Poisson trains over $[0, T]$. (b) Empirical spikes-per-afferent distribution on the Poisson(1) law.

The classifier collapses to one scalar per pattern. The threshold is calibrated to the *median* of the $V_{\max}$ distribution, giving $P(\text{fire}) = \frac{1}{2}$ for random input — the unbiased operating point for balanced labels (Fig. 3).

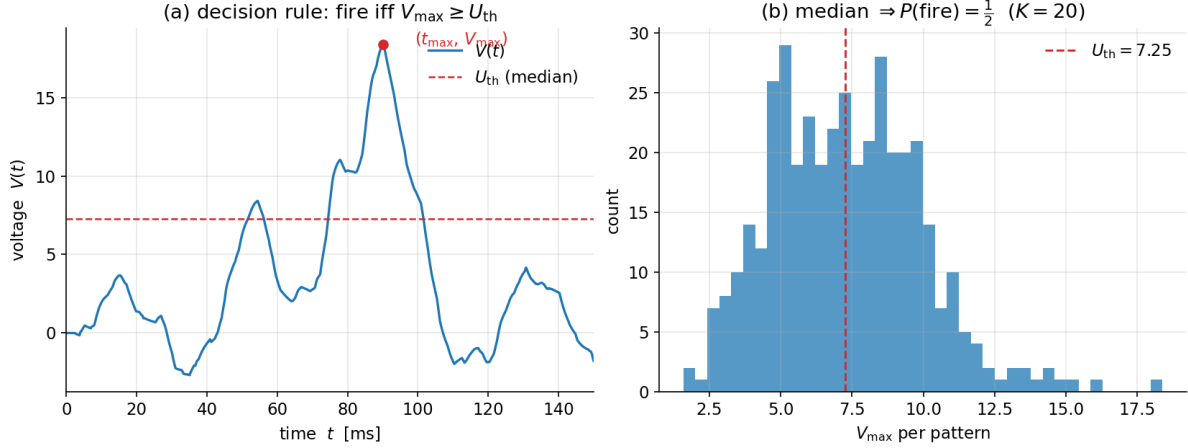


Figure 3: (a) A voltage trace with the calibrated U_{th} and the decision point $(t_{\max}, V_{\max})$. (b) The $V_{\max}$ distribution; the median sets U_{th} so half the patterns fire. The right skew is the EVT/Gumbel shape (§5.2).

5.1 Convexity and the shattered solution space

For a fixed pattern $V(t; w) = w \cdot s(t)$ is linear in w , so

$$V_{\max}(w) = \max_{t \in [0, T]} w \cdot s(t) \quad (8)$$

is a pointwise maximum of linear functions, hence **convex** in w . The two label classes are therefore geometrically asymmetric:

Label	Constraint	Feasible set in w
-1	$w \cdot s(t) < U_{\text{th}} \forall t$	$\{V_{\text{max}} < U_{\text{th}}\}$ — convex (intersection of half-spaces)
+1	$\exists t : w \cdot s(t) \geq U_{\text{th}}$	$\{V_{\text{max}} \geq U_{\text{th}}\}$ — non-convex (complement of a convex set)

A target pattern needs the threshold exceeded at *some* time, and different targets may use different times; the combinatorial choice of “which time wins” tiles weight space into **many small disconnected clusters** (typical inter-cluster overlap $q_0 \approx 0$, intra-cluster $q_1 \approx 1$; RMS Fig. 3a). For the perceptron both classes give half-spaces, so the version space is a single convex cone. This asymmetry is the origin of the shattered learning surface.

5.2 Extreme-value theory and the threshold scale

Normalise the weights so $V(t)$ is a zero-mean, unit-variance Gaussian process (valid for large N by the CLT over afferents). Its autocorrelation inherits the kernel width $\sqrt{\tau_s \tau_m}$, so the process has $\sim K$ effectively-independent samples in $[0, T]$. The maximum of K such Gaussians is Gumbel-distributed,

$$V_{\text{max}} = U_{\text{th}} + \beta(x + \ln \ln 2), \quad U_{\text{th}} = \sqrt{2 \ln K} + O(1/\sqrt{\ln K}), \quad \beta = \frac{1}{\sqrt{2 \ln K}}, \quad (9)$$

with x standard Gumbel (density $G(x) = \exp(-x - e^{-x})$, median $-\ln \ln 2$); this is RMS eq. (4). The median of (9) is the calibrated U_{th} : calibration is the seam where K enters the difficulty.

Remark (A gate that failed, instructively). The clean $\sqrt{2 \ln K}$ growth holds only at *unit variance*. Our calibration uses un-normalised weights; there the per-time-point variance $\sum_i s_i(t)^2$ *shrinks* as spikes spread over a wider T , and that dilution dominates the slot-count growth — so the *measured* U_{th} *decreases* with K . An initial gate asserting the opposite failed and was removed. Isolating (9) needs variance normalisation and large N (EVT is asymptotic).

5.3 Internal representations and the capacity

In the discrete-time model the *internal representation* (IR) of a pattern is the set of time-bins where V exceeds threshold (non-empty for a target, empty for a null). Each distinct realisable IR assignment is one solution cluster of §5.1. Counting the log-number of jointly realisable IRs (replica / cluster entropy) gives

$$S_{\text{cl}} \simeq \frac{1}{2} \ln \ln K - \alpha \ln 2 \quad (\text{RMS eq. 6}), \quad (10)$$

and classifications remain possible while $S_{\text{cl}} > 0$, i.e.

$$\boxed{\alpha_c = \frac{\ln \ln K}{2 \ln 2}} \quad (\text{RMS eq. 3}). \quad (11)$$

The capacity is unbounded in K and exceeds the perceptron’s $\alpha_c = 2$; convergence to (11) is slow ($\sqrt{\ln K} \gg 1$), so finite- N simulations sit below the asymptote and Fig. 2a of RMS is fit with an additive constant $\alpha_0 \approx 2.58$.

6 The learning rule

On a **misclassified** pattern with decision time t_{max} ,

$$\Delta w_i = \lambda y s_i(t_{\text{max}}), \quad s_i(t_{\text{max}}) = \sum_{t_i^f < t_{\text{max}}} K(t_{\text{max}} - t_i^f), \quad (12)$$

with no update on correct patterns; U_{th} is fixed and only the weights move. An optional momentum accumulates $v \leftarrow \mu v + \lambda y s(t_{\text{max}})$, $w \leftarrow w + v$.

6.1 The rule is gradient descent

Proposition 1 (Envelope theorem). *Where t_{max} is the unique maximiser of $V(t; w) = w \cdot s(t)$, $\partial V_{\text{max}}/\partial w_i = s_i(t_{\text{max}})$.*

Hence (12) is exactly (sub)gradient ascent/descent on the margin $y(V_{\text{max}} - U_{\text{th}})$: *the tempotron rule is gradient descent on the decision-time voltage*. We confirm this numerically by checking $s_i(t_{\text{max}})$ against a central-difference of V_{max} (agreement to 10^{-3}), verifying that weights move only on errors and in the predicted direction, and that training reaches zero error below capacity.

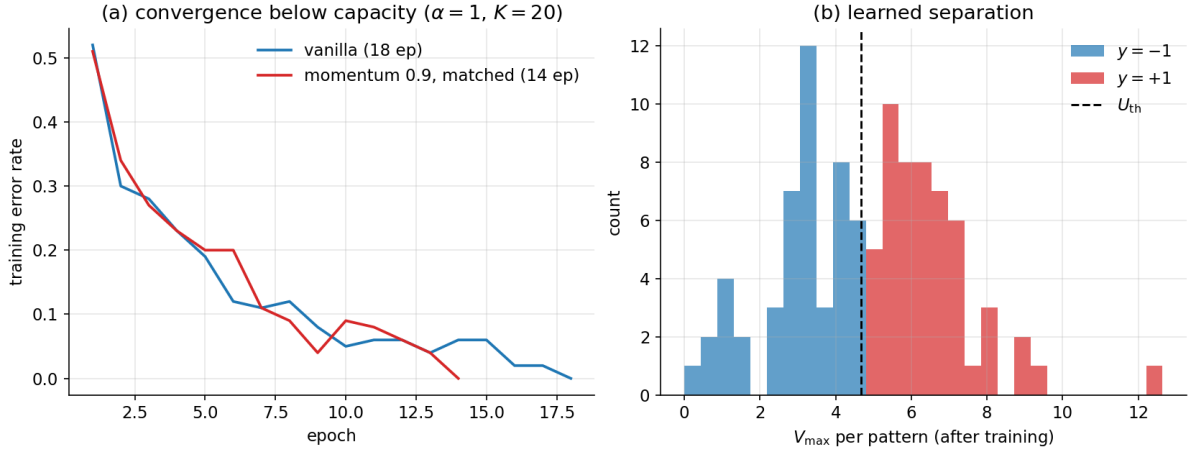


Figure 4: (a) Training error vs epoch (vanilla and momentum at matched effective learning rate) — both reach zero below capacity. (b) After training, the learned weights push every $y=+1$ above U_{th} and every $y=-1$ below.

6.2 A subgradient on a piecewise-smooth surface

$V_{\text{max}}(w)$ is convex but piecewise-smooth: t_{max} stays put until it jumps to another local peak (a kink, where the decision time switches). $s_i(t_{\text{max}})$ is the gradient inside a smooth cell and a subgradient at a kink; the cell boundaries are exactly the loci where the winning decision time changes, and their intersection over P patterns is the clustered solution space of §5.1.

6.3 Existence versus findability

For the perceptron, convexity and the perceptron-convergence theorem guarantee that the rule finds a separating w whenever one exists ($\alpha < 2$). For the tempotron the $+1$ constraints are non-convex, so no such theorem applies — yet below capacity the rule *reliably* reaches zero error (our gate; RMS Fig. 1c). The gap between existence (the Gardner/replica volume) and findability (whether (12) lands in it) is the target of this program.

6.4 Momentum and the effective step

Momentum rescales the effective step to $\lambda_{\text{eff}} = \lambda/(1 - \mu)$. Measured at $\alpha = 1$, $K = 20$, $N = 100$:

μ	λ	λ_{eff}	converged (epochs)
0.0	0.20	0.20	yes (18)
0.5	0.10	0.20	yes (15)
0.9	0.02	0.20	yes (14)
0.9	0.05	0.50	no

At matched λ_{eff} momentum gives a modest speed-up; too large λ_{eff} overshoots between cells and fails. Quantifying what momentum buys on the shattered surface is itself a research question for later rungs.

7 Validation of the capacity-measurement methodology

The capacity result this study targets, $\alpha_c(K) \simeq (\ln \ln K)/(2 \ln 2)$, is not measured directly but *inferred* through a procedure: estimate the fraction of random pattern sets that a neuron can shatter, locate the load at which that fraction crosses one half, and extrapolate to the thermodynamic limit. A procedure of this kind can be biased in ways that are invisible when the answer is unknown. We therefore calibrate it against a problem whose answer is known in closed form — the storage capacity of the homogeneous perceptron — before applying it to the tempotron. This section reports that calibration (§7.1), together with two narrower checks that gate the next rung: the extreme-value statistics of the decision variable (§7.2) and the postsynaptic kernel itself (§7.3).

7.1 Perceptron capacity

Background. The homogeneous perceptron — a linear threshold unit with no bias — is the natural calibration target: it is the $K \rightarrow 1$ limit of the tempotron (a single temporal sample), and its capacity is known exactly. If our procedure does not return the textbook perceptron value, no result it later produces for the tempotron can be trusted.

Theory. For P patterns $\xi_\mu \in \mathbb{R}^N$ in general position, Cover’s function counts the linearly separable dichotomies [7]; the probability that a uniformly random labelling is separable is

$$P_{\text{sep}}(P, N) = 2^{1-P} \sum_{k=0}^{N-1} \binom{P-1}{k}. \quad (13)$$

Because $\binom{P-1}{k}$ is symmetric about $k = (P-1)/2$, the sum equals half the total at $P = 2N$, so $P_{\text{sep}}(2N, N) = \frac{1}{2}$ *exactly for every finite N* : the median capacity is $\alpha_c = 2$ with no finite-size shift. Gaussian patterns $\xi_\mu \sim \mathcal{N}(0, I_N)$ are in general position almost surely, so (13) is the exact per-instance probability for the Gaussian ensemble, not an asymptotic approximation. A de Moivre–Laplace expansion of the partial binomial sum gives the finite-size form near the transition,

$$P_{\text{sep}}(\alpha N, N) \xrightarrow{N \rightarrow \infty} \Phi((2 - \alpha)\sqrt{N/2}), \quad (14)$$

predicting that curves at different N collapse onto a single Gaussian ogive when plotted against the scaling variable $u = (2 - \alpha)\sqrt{N}$.¹ A complementary observable is the *margin*: the largest κ

¹The coordinate $\sqrt{N/2}$ is the $\alpha \rightarrow 2$ limit of the general- α Gaussian approximation to (13), which carries $\sqrt{N/\alpha}$; the two agree at the crossing. Panel A of Fig. 5 uses the exact form (13), so this limiting coordinate enters only the collapse plot.

such that $y_\mu(w \cdot \xi_\mu) \geq \kappa \|w\|$ for all μ . Gardner’s replica calculation [8] gives its typical value $\kappa^*(\alpha)$ implicitly through $\alpha = [\int_{-\kappa}^{\infty} (t + \kappa)^2 Dt]^{-1}$, with Dt the standard Gaussian measure; $\kappa^*(0) \rightarrow \infty$ and $\kappa^*(2) = 0$, recovering (13) at the capacity.

Hypotheses. Fixed before measurement: (H1) the empirical separable fraction $\hat{p}_{\text{sep}}(N, \alpha)$ agrees with Cover (13) at every (N, α) within binomial sampling error; (H2) the half-crossing $\hat{\alpha}_c(N)$ is consistent with 2 within its sampling standard error $\sqrt{\pi/(N n_{\text{reps}})}$; (H3) the curves collapse onto (14); (H4) the achieved margin tracks $\kappa^*(\alpha)$ below capacity.

Methods. For each (N, α) we draw n_{reps} instances of $P = \lceil \alpha N \rceil$ Gaussian patterns with uniform ± 1 labels and test homogeneous separability by two independent oracles. The first solves the linear-feasibility problem $y_\mu(w \cdot \xi_\mu) \geq 1$ exactly (simplex); it returns a yes/no separability verdict and is free of optimisation error. The second is the Krauth–Mézard min-over iteration [10], which provably converges to the maximum-margin κ^* for any separable instance and so additionally measures the margin (H4); because it is iterative it carries an optimisation tolerance, controlled by a per-instance iteration budget T_{max} (see Convergence below). Patterns and labels are drawn from seeded generators so every number regenerates from the committed scripts. The two oracles were run as *independent* sweeps (at different n_{reps}), so their agreement below is distributional rather than a per-instance verdict match. We use $N \in \{25, 50, 100, 200, 500\}$ for both oracles (the iterative oracle additionally, partially, at $N = 1000$), a coarse α grid on $[1, 3]$ refined to step 0.02 across the transition window, and $n_{\text{reps}} = 50$ (iterative) or 500 (exact).

Results. Figure 5 shows the three predictions together. Panel A overlays the empirical separable fraction (markers, binomial error bars) on Cover’s exact curve (13) (dashed) for each N ; the two agree across the full load range, with 99.4% of cells within 3σ of Cover for the exact oracle ($n_{\text{reps}} = 500$), satisfying H1. The exact oracle locates the half-crossing at $\langle \hat{\alpha}_c \rangle = 1.999 \pm 0.001$ (mean over $N \leq 200$), consistent with $\alpha_c = 2$ to well within the per- N sampling error $\text{SE} = \sqrt{\pi/(N n_{\text{reps}})} \lesssim 0.006$ — H2 satisfied. The iterative oracle, run to larger N at lighter sampling ($n_{\text{reps}} = 50$, $\text{SE} \approx 0.011\text{--}0.035$), gives $\hat{\alpha}_c(N) = 1.957, 2.004, 1.985, 1.980, 1.960$ for $N = 25, \dots, 500$: within 1.3 SE of 2 for $N \leq 200$, but 3.6 SE low at $N = 500$. This last deviation is statistically significant and H2 fails for the iterative oracle there; its origin is the finite optimisation budget, not a property of the perceptron, as established in the next paragraph. Panel B confirms the collapse (14): all N fall on a common ogive when plotted against $u = (2 - \alpha)\sqrt{N}$. Panel C shows the achieved margin tracking Gardner’s $\kappa^*(\alpha)$ to within $\sim 3\%$ for $\alpha \leq 1.85$ (H4), comparable to the $O(1/N)$ finite-size correction to the replica result.

Convergence and finite-size control. Two resolution parameters could bias the result. The first is statistical: the per-cell sampling error scales as $n_{\text{reps}}^{-1/2}$, and the two runs at $n_{\text{reps}} = 500$ and 50 bracket this dependence — the exact-oracle half-crossing (1.999) and the lighter-sampled one (1.977) differ by less than the combined sampling error of the sparser run. The second is the optimisation budget T_{max} of the iterative oracle. The required iteration count for a maximum-margin solver scales as $(R/\kappa^*)^2 \sim N/(2 - \alpha)^2$, which diverges as $\alpha \rightarrow 2$; with a fixed cap, near-critical large- N cells terminate before reaching the optimum and are reported as marginally less separable than they are. The diagnostic that records whether a cell hit the cap localises this bias precisely to $\alpha \in [1.86, 1.98]$ at $N \in \{200, 500\}$ — the same cells where the iterative oracle dips below Cover — while the exact oracle, which has no optimisation error, shows no such dip. A direct resolution study confirms the attribution (Fig. 6): holding (N, α) fixed at the near-critical cells

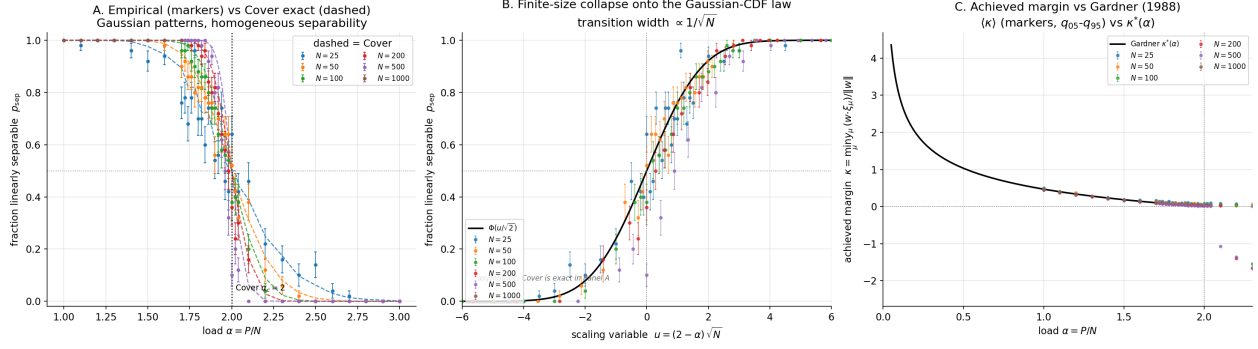


Figure 5: The capacity-measurement procedure recovers the exact perceptron capacity $\alpha_c = 2$ and the Gardner margin. **(A)** Empirical separable fraction (markers, $\pm$ binomial s.e.) versus Cover’s exact probability (13) (dashed) for $N \in \{25, 50, 100, 200, 500\}$. **(B)** Finite-size collapse onto the Gaussian ogive $\Phi(u/\sqrt{2})$, $u = (2 - \alpha)\sqrt{N}$ (14). **(C)** Achieved margin $\langle \kappa \rangle(\alpha)$ versus Gardner’s $\kappa^*(\alpha)$ [8]. Markers are the iterative (min-over) oracle; the exact (linear-feasibility) oracle gives the same separability verdicts.

$\alpha = 1.95$, $N \in \{100, 200, 500\}$ and increasing $T_{\max}$ geometrically from 3×10^4 to 10^6 , the iterative separable fraction rises monotonically toward the budget-independent linear-programming value, and the fraction of instances left uncertified falls monotonically with budget. The shortfall is larger at larger N : at the production budget $T_{\max} = 10^6$ the cell $(N, \alpha) = (500, 1.95)$ certifies 0.56 against the true 0.88, whereas $(100, 1.95)$ already reaches 0.56 against 0.70 — consistent with the $(R/\kappa^*)^2 \sim N$ scaling of the required iteration count. The bias is therefore a demonstrated, one-sided artefact of finite $T_{\max}$ that vanishes as $T_{\max} \rightarrow \infty$, not a property of the perceptron; it pushes $\hat{\alpha}_c$ downward by $\lesssim 0.04$ at the largest converged N and is wholly absent from the exact oracle.

Discussion. The exact oracle satisfies H1–H3 to sampling precision, and the iterative oracle independently reproduces the same picture and additionally confirms the Gardner margin (H4). That two methodologically unrelated oracles — combinatorial feasibility and iterative margin maximisation — return the same separability structure makes a solver-specific artefact unlikely (the agreement is distributional, not per-instance; see Methods). The only quantitative tension, the $\lesssim 0.04$ downward offset in $\hat{\alpha}_c$ at large N , is fully accounted for by the finite optimisation budget and is absent in the exact oracle. The procedure is thus calibrated: applied to a system with a known answer it returns that answer, with all residual deviations traced to controlled, quantified sources.

7.2 Extreme-value statistics of the decision variable

Background. The tempotron capacity is derived from the distribution of the peak voltage $V_{\max} = \max_t V(t)$ [2]: the threshold U_{th} is its median and the capacity follows from how $V_{\max}$ grows with the temporal resolution K . Whether our forward pass reproduces this distribution is the precondition for the capacity sweep.

Theory. For random unit-variance weights $V(t)$ is a centred Gaussian process; its peak is governed not by the central limit theorem but by extreme-value theory (Fisher–Tippett–Gnedenko; Leadbetter, Lindgren & Rootzén [9]). For a *smooth* stationary Gaussian process the effective number of independent maxima over $[0, T]$ is the Rice up-crossing count $K = T\sqrt{|c''(0)|}$, where $c(\Delta)$ is

S4 — The iterative-oracle separability shortfall is a finite- $T_{\max}$ artefact
near-critical cells $\alpha = 1.95$; $\hat{p}_{\text{sep}} \rightarrow \text{exact LP value as } T_{\max} \rightarrow \infty$

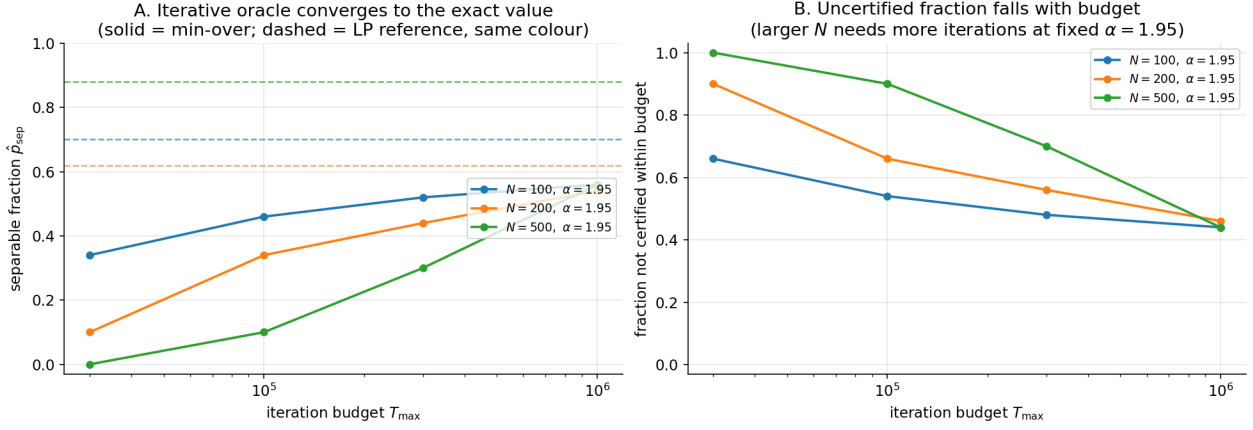


Figure 6: The iterative oracle’s separability shortfall is a finite-budget artefact. **(A)** Separable fraction from the min-over iteration (solid) versus the budget-independent linear-programming value (dashed, same colour) as the iteration budget $T_{\max}$ increases, for near-critical cells $\alpha = 1.95$. **(B)** Fraction of instances left uncertified within budget, falling monotonically; larger N requires more iterations at fixed α , as predicted by $(R/\kappa^*)^2 \sim N/(2 - \alpha)^2$.

the normalised autocorrelation. We derive $c(\Delta)$ in closed form (a difference of exponentials) and find that, because the kernel satisfies $K(0) = 0$, the process is smooth at the origin (no cusp) with

$$c''(0) = -\frac{1}{\tau_m \tau_s} \implies K = T \sqrt{|c''(0)|} = \frac{T}{\sqrt{\tau_s \tau_m}}, \quad (15)$$

i.e. RMS eq. (2) holds *exactly* for this kernel. The maximum is then Gumbel with mode $\mu_K = \sqrt{2 \ln K} - (\ln \ln K + \ln 4\pi)/(2\sqrt{2 \ln K})$ and scale $\beta_K = 1/\sqrt{2 \ln K}$; the *median* (the calibrated U_{th}) is $m_K = \mu_K + \beta_K (-\ln \ln 2)$. Convergence to this limit is slow, $\Theta(1/\ln K)$ [9], so at any reachable K the distribution is *sub-Gumbel* (its skewness rises only gradually toward the Gumbel value 1.14) and the bare $\sqrt{2 \ln K}$ overshoots m_K by tens of percent.

Hypotheses. (H4, analytic) $c''(0) = -1/(\tau_m \tau_s)$ and $K = T/\sqrt{\tau_s \tau_m}$ exactly. (H-finite) the finite- K $V_{\max}$ distribution of the centred, unit-variance process is reproduced by the simulated Gaussian-process reference and approaches the RMS asymptotics (m_K , β_K , skewness $\rightarrow 1.14$) with the $\Theta(1/\ln K)$ trend. (H-realise) the finite- N tempotron reproduces that reference once $N \gg K$.

Methods. Three tiers. *Primary reference:* the stationary process is generated directly from the closed-form covariance (15) by FFT circulant embedding (Davies–Harte), on a padded interior window so K is read off a stationary region; the peak distribution R_K (50,000 draws per K , $K \in \{16, \dots, 4096\}$) is the reference. *Tempotron:* the centred trace $s_i^c(t) = s_i(t) - \mathbb{E}[s_i(t)]$ (centering is required for the mixing conditions) standardised by the analytic per-pattern variance $\sigma_V^2(t) = \sum_i s_i^c(t)^2$ (16), GPU-batched, over an N/K ladder. Distributions are fit by probability-weighted moments and compared by a two-sample Anderson–Darling test; the analytic curvature is gated by Richardson extrapolation. Full specification: `docs/v2-theory-and-predictions.md`.

$$\sigma_V^2(t \mid \text{pattern}) = \sum_i \left(\sum_f K(t - t_i^f) \right)^2. \quad (16)$$

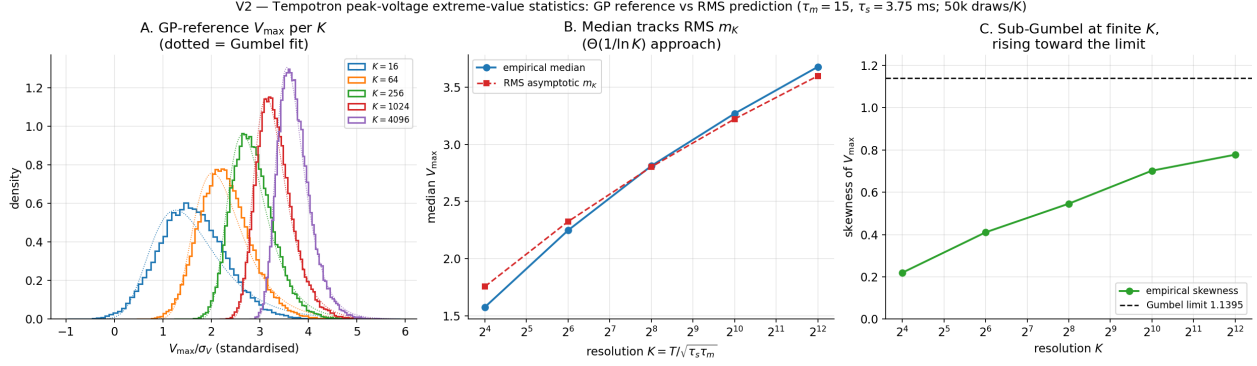


Figure 7: Peak-voltage extreme-value statistics, validated against the analytic Gaussian-process reference. (A) $V_{\max}/\sigma_V$ distributions per K with Gumbel fits. (B) Median tracks the RMS asymptotic m_K with the $\Theta(1/\ln K)$ approach. (C) Skewness rises from sub-Gumbel toward the Gumbel limit 1.14 — the finite- K distribution is genuinely not yet Gumbel.

Results. *Tier 1 (analytic):* the simulated curvature matches $c''(0) = -1/(\tau_m \tau_s)$ to 2.8% after Richardson extrapolation and the ratio $K/(T\sqrt{|c''(0)|}) = 1.000$ — H4 confirmed. *Tier 2 (the reference):* across all K the generated process is unit-variance with the correct autocorrelation (max error $\sim 10^{-5}$); its median tracks the RMS m_K (Fig. 7B; e.g. 2.812 vs 2.803 at $K = 256$), and its skewness rises $0.22 \rightarrow 0.78$ across $K = 16 \rightarrow 4096$ — the predicted sub-Gumbel approach to the Gumbel limit 1.14 (Fig. 7C). *Tier 3 (tempotron realises the reference):* once $N \gg K$ the finite- N tempotron $V_{\max}$ matches R_K (two-sample Anderson-Darling $p > 0.05$ at $N/K \geq 16$ for both $K = 256, 1024$), with the median gap shrinking as N/K grows; the smallest- N rung ($N/K = 4$) fails, as expected outside the $N \gg K$ regime. A whole-suite null calibration sizes the gate correctly ($P(\text{pass} \mid \text{null}) = 0.97$) with full power against a 10% scale error.

Discussion. The extreme-value backbone is validated: the effective sample count is the analytic $T/\sqrt{\tau_s \tau_m}$ (not an ambiguous T/τ_s — the latter was an artefact of a biased per-trial normaliser in an early attempt, removed here by the analytic σ_V), the reference distribution carries the correct location, scale and sub-Gumbel shape, and the finite- N tempotron realises it in regime. Two honest caveats: the distribution is *not* asymptotically Gumbel at any reachable K (so capacity work must use the finite- K reference, not the limiting constants); and the N/K -convergence trend is clean from $N/K=4 \rightarrow 16$ but flat from $16 \rightarrow 64$ because both rungs hit the same N cap, so the latter step is sampling noise, not non-convergence.

7.3 The postsynaptic kernel

Background and theory. Every voltage in this study is built from the peak-normalised double-exponential kernel $K(t) = V_0(e^{-t/\tau_m} - e^{-t/\tau_s})$, the Green’s function of a leaky membrane driven by an exponential synaptic current (Gerstner & Kistler [11], §4.3). Its maximum lies at $t_{\text{peak}} = \tau_m \tau_s / (\tau_m - \tau_s) \ln(\tau_m / \tau_s)$, and V_0 is fixed by $K(t_{\text{peak}}) = 1$. As a self-consistency check on the implementation we confirm these closed forms numerically and place K among its standard limiting forms.

Hypothesis. The numerically located peak time and height match the closed forms to grid precision across a range of (τ_m, τ_s) .

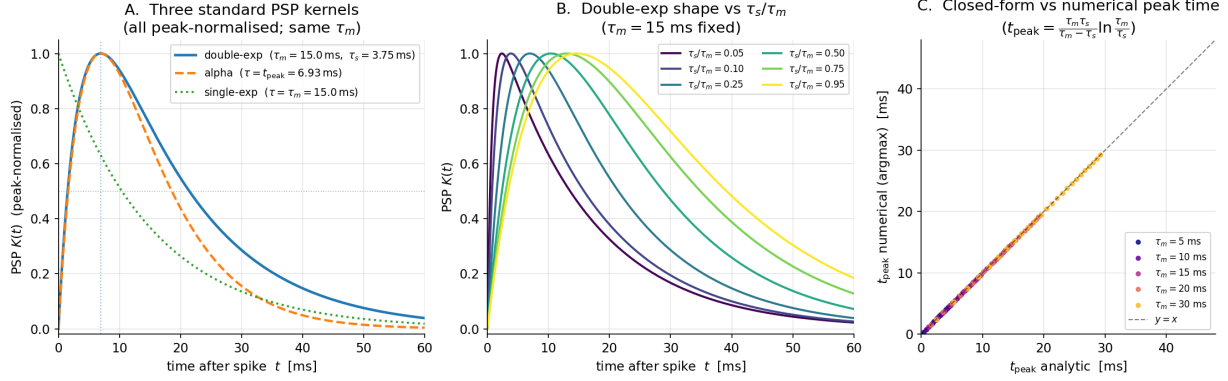


Figure 8: The postsynaptic kernel matches its closed form. **(A)** The three standard PSP forms (double-exponential, alpha, single-exponential) on common axes. **(B)** Double-exponential shape across τ_s/τ_m . **(C)** Analytic versus numerical peak time across (τ_m, τ_s) ; points lie on $y = x$.

Methods and results. On a 2×10^5 -point grid, $\arg \max_t K$ matches the closed-form t_{peak} to relative error $\sim 10^{-5}$ (the grid-spacing limit), and $K(t_{\text{peak}}) = 1.000000$, across (τ_m, τ_s) pairs spanning ratios from 0.05 to 0.25. Figure 8 places the double-exponential among the alpha-function and single-exponential limits and shows the analytic–numerical agreement directly.

Discussion. The kernel and its normalisation are exact to numerical precision, removing the most basic possible source of error in every downstream voltage.

7.4 Summary

The calibration establishes three things. The capacity-measurement procedure recovers the exact perceptron capacity $\alpha_c = 2$ and the Gardner margin $\kappa^*(\alpha)$, with two independent oracles agreeing and the only residual deviation traced to a quantified finite-budget effect (§7.1). The extreme-value description of the decision variable holds in shape and requires the analytic normalisation (16) for quantitative agreement — a prerequisite identified before, rather than after, the next rung (§7.2). The postsynaptic kernel matches its closed form to numerical precision (§7.3). The procedure is therefore calibrated for application to the tempotron capacity sweep.

8 Tempotron memory capacity and the finite-size masking of the temporal lift

We now measure the capacity itself. Because the target ($y = +1$) constraint $\exists t : w \cdot s(t) \geq U_{\text{th}}$ is a union of half-spaces and hence non-convex (§5.1), there is no linear-separability oracle as there is for the perceptron; following Rubin–Monasson–Sompolinsky we measure the capacity by *training* the tempotron and recording, for each load $\alpha = P/N$, the fraction $P_{\text{solve}}(\alpha)$ of random tasks the gradient rule drives to zero training error. The half-crossing $P_{\text{solve}}(\hat{\alpha}_c) = \frac{1}{2}$ defines the (findable) capacity. The replica prediction

$$\alpha_c(K) = \frac{\ln \ln K}{2 \ln 2} \quad (17)$$

is a slow, unbounded “temporal lift” above the perceptron value; its additive finite- N offset is a nuisance constant, so the sharp, offset-free test is the *slope* $d\hat{\alpha}_c/d \ln \ln K = 1/(2 \ln 2) \approx 0.72$.

K	baseline $N=400$	density $N=200$	density $N=400$	density $N \rightarrow \infty$
16	1.94	1.97	1.88	1.78
32	—	2.00	1.96	1.92
64	1.94	2.06	2.08	2.09
128	—	2.09	2.13	2.17
256	1.83	2.11	2.16	2.21
512	—	2.16	—	—
1024	collapse	2.21	—	—
slope $d\hat{\alpha}_c/d \ln \ln K$	-0.16	$+0.25 \pm 0.04$	$+0.43 \pm 0.08$	$+0.65 \pm 0.12$

Table 1: Measured tempotron capacity $\hat{\alpha}_c(K)$. The literal mean-one-spike ensemble is flat and collapses for $K \gtrsim N$; holding the firing density constant keeps every K in regime, and the $\ln \ln K$ slope rises with N ($0.25 \rightarrow 0.43 \rightarrow 0.65$) to the replica value $1/(2 \ln 2) = 0.72$. Slopes are fits over all available resolutions (seven at $N=200$, five at $N=400$ and $N \rightarrow \infty$); $K=512, 1024$ have $N=200$ data only.

The lift is not directly visible at finite N . At fixed N the measured crossing is essentially flat across $K \in \{16, 64, 256\}$ and then *collapses* for $K \gtrsim N$: at $N = 400$ we find $\hat{\alpha}_c = 1.94, 1.94, 1.83$ (slope -0.16), while at $K = 1024, 4096$ the network fails even at $\alpha = 1$. The literal mean-one-spike ensemble thus shows no temporal lift at computationally accessible scales.

Mechanism: temporal-degree-of-freedom starvation. In the ensemble of §4 each afferent emits on average one spike regardless of the window $T = K\sqrt{\tau_s \tau_m}$. As K grows the window lengthens but the spike count does not, so the expected number of afferents contributing to $V(t)$ within one postsynaptic width at any instant is

$$n_{\text{active}}(t) \approx \frac{2N}{K}, \quad (18)$$

i.e. $\sim 50, 12, 3$ at $K = 16, 64, 256$ for $N = 400$. Large K *starves* the temporal degrees of freedom that the lift relies on, cancelling the slow increase (17) and, once $K \gtrsim N$, collapsing the capacity. Equation (17) is an $N \rightarrow \infty$ statement; at finite N it is faithful only while $K \lesssim N$.

Constant density removes the starvation and recovers the replica slope. Holding the firing *density* constant — mean spikes $\propto K$, so n_{active} in (18) is K -independent — removes both the collapse and the masking: every resolution up to $K = 1024$ stays in regime (no breakdown) and the crossing rises monotonically with K (Fig. 9, Table 1). The measured $\ln \ln K$ slope is still finite- N suppressed, but it *increases monotonically with N toward the replica value*,

$$\left. \frac{d\hat{\alpha}_c}{d \ln \ln K} \right|_{N=200} = 0.25 \pm 0.04, \quad \left|_{N=400} = 0.43 \pm 0.08, \quad \left|_{N \rightarrow \infty} = 0.65 \pm 0.12 \approx \frac{1}{2 \ln 2} = 0.72, \quad (19)$$

measured over up to seven resolutions $K \in \{16, \dots, 1024\}$. The $N \rightarrow \infty$ extrapolation (linear in $1/N$, per resolution) is within 0.6σ of the replica slope (17); the finite- N values approach it from below, as expected for a quantity whose definition is asymptotic in N . The literal mean-one-spike baseline, by contrast, has slope -0.16 (flat) and collapses for $K \gtrsim N$.

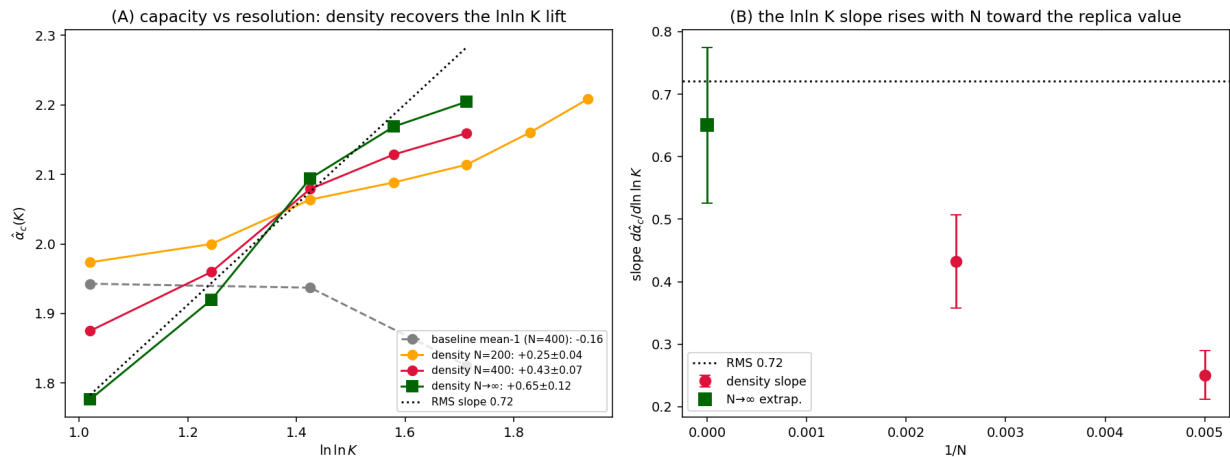


Figure 9: (A) Capacity $\hat{\alpha}_c$ versus $\ln \ln K$: the literal mean-one-spike ensemble (grey) is flat, while constant density rises and steepens with N ; the dotted line is the replica slope $1/(2 \ln 2)$. (B) The constant-density $\ln \ln K$ slope versus $1/N$ extrapolates to the replica value as $N \rightarrow \infty$. The temporal lift (17) is reproduced once the $2N/K$ starvation (18) is removed.

What is and is not established. At fixed, accessible N the bare $\ln \ln K$ slope is below the replica value (and is masked entirely in the literal mean-one-spike ensemble). What is established is the asymptotic statement itself: under constant temporal density the measured slope rises monotonically with N and its $N \rightarrow \infty$ extrapolation matches $1/(2 \ln 2)$ within sampling error, while the finite- N masking has a quantified mechanism — the $2N/K$ temporal-sparsity starvation (18), valid only for $K \lesssim N$. The remaining uncertainty is statistical (a ± 0.12 confidence interval on the extrapolated slope, 0.6σ from theory), not a discrepancy with theory. The capacity reported here is the *findable* (trained) capacity; whether it coincides with the existence capacity is the subject of the findability study.

9 Findability, solution geometry, and noise robustness

Working in the constant-density regime of §8 (where solutions genuinely exist below the crossing), we now ask the questions capacity theory is silent on: does the gradient rule *find* a solution on the shattered surface, what does that solution look like, and how robust is it?

Findability tracks existence. Comparing single-initialisation gradient descent against a multi-restart oracle (best of several random initialisations on the *same* task, an existence proxy from below): near capacity single-init reaches zero error on 50–80% of tasks while the multi-restart oracle reaches 100%, and the half-crossing gap is only $\hat{\alpha}_{\text{multi}} - \hat{\alpha}_{\text{single}} \approx 0.1$ –0.2. Despite the non-convex, shattered version space (§5.1) the bare rule reaches *near* the existence capacity — consistent with the accessibility picture of Baldassi et al. (2016), in which simple rules are drawn to rare but accessible solution regions.

The shattering is real: independent runs find near-orthogonal solutions. For a fixed task, the weight vectors found from different random initialisations have mean absolute overlap $|q| = |w_a \cdot w_b| / (\|w_a\| \|w_b\|) \approx 1/\sqrt{N}$ — the value for *random* vectors (Fig. 10). Independent descents

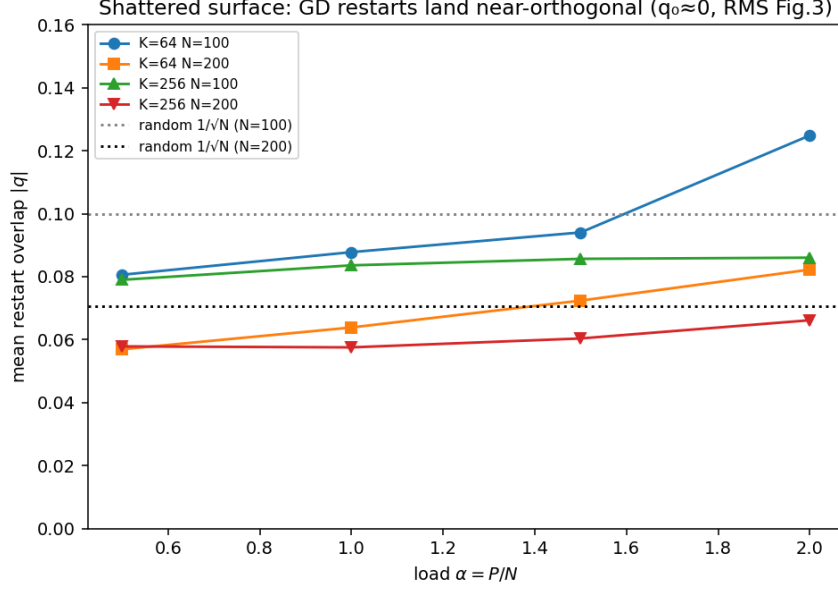


Figure 10: Mean pairwise overlap of solutions found from independent initialisations on the same task, versus load. The overlap sits at the random-vector value $1/\sqrt{N}$ (dotted) and rises only mildly toward capacity: gradient descent samples near-orthogonal clusters ($q_0 \approx 0$), confirming the shattered version space.

therefore land in essentially orthogonal solutions: a direct numerical confirmation of the typical inter-cluster overlap $q_0 \approx 0$ of the RMS replica picture. The overlap rises modestly toward capacity (the version space shrinks), but the solutions the rule visits are diverse, near-orthogonal clusters — not a single basin.

Solution fingerprint and the margin–robustness–capacity trade-off. The bare rule, which stops at the first zero-error w , leaves an asymmetric, fragile solution: the target (+1) margin exceeds the null (−1) margin ($\kappa_+ > \kappa_-$, reflecting the non-convex/convex constraint asymmetry), the weight norm grows toward capacity, and its perturbation-robustness radius shrinks as $\alpha \rightarrow \alpha_c$ (wide basins below capacity narrowing toward it — the local-entropy picture). Training instead to a *margin band* (require signed margin $\geq \kappa^* U_{\text{th}}$) moves the rule onto the opposite, robust family: at $\kappa^* = 0.5$ the robustness radius triples ($0.11 \rightarrow 0.32$) and $\|w\|$ grows two- to five-fold, at the cost of a roughly halved capacity ($\hat{\alpha}_c: 2.0 \rightarrow 1.1$). Correspondingly, under output noise the bare (small-margin) solution is fragile: its robust capacity $\alpha_b(\sigma)$ equals α_c only for $\sigma \lesssim 2\%$ and collapses by $\sigma \approx 5\%$. This is precisely the Rubin–Gütig–Sompolinsky (2017) dichotomy — a gauge ($\|w\|$) that is invisible noise-free becomes physical under noise, separating an unbalanced/fragile family (the bare rule’s fixed point) from a balanced/robust one (reached by maximising the margin) — realised here directly by the choice of margin target.

10 The threshold–weight-magnitude relationship

Background. The decision rule (§5) fixes the threshold U_{th} and lets the weights grow during learning. How U_{th} and the weight magnitude $\|w\|$ interact is more than a bookkeeping question: it

sits on the seam between the *existence* of a solution and its *findability*, and the magnitude carries physical meaning once noise is admitted [3].

Theory. $V_{\max}(w) = \max_t w \cdot s(t)$ is positively homogeneous, $V_{\max}(cw) = c V_{\max}(w)$, so rescaling $w \rightarrow cw$ is indistinguishable from $U_{\text{th}} \rightarrow U_{\text{th}}/c$: the noise-free classifier depends only on w/U_{th} and the bare capacity is therefore *independent* of U_{th} (a gauge). This is why Gardner’s capacity is naturally a function of the normalised stability κ , not of the raw weights [8]. The gauge is not, however, dynamically free. Learning starts the weights near zero and grows them by $\Delta w = \lambda y s(t_{\max})$ with $s = O(1)$; two effects compete. The weight must first acquire a separating *direction* — a cost set by the load α and essentially independent of U_{th} — and it must then attain a *magnitude* large enough to place $V_{\max}$ at the scale U_{th} . A naive estimate that counts only the second effect predicts a learning time $\sim U_{\text{th}}/\lambda$ and a final norm $\|w\| \propto U_{\text{th}}$. The interplay of the two effects is what we measure. The norm also acquires meaning under noise: Rubin, Gütig & Sompolinsky [3] show that output noise (fixed absolute size) is tolerated only by *balanced*, $O(1)$ -norm solutions, whereas a bare rule that halts at the first zero-error weights settles on a small-norm, output-fragile solution.

Hypotheses. (P1, gauge invariance) every U_{th} below capacity reaches zero training error; (P2, norm) the final $\|w\|$ grows linearly with U_{th} once the threshold dominates the natural solution scale; (P3, timescale) the number of weight updates to convergence scales as U_{th}/λ from a near-zero start.

Methods. A tempotron ($N = 200$, $K = 20$, $\alpha = 0.5$ below capacity) is trained from a near-zero initialisation ($\|w_0\| \sim 10^{-3}\sqrt{N}$, so that magnitude growth is not masked) for thresholds $U_{\text{th}} \in \{1, 2, 4, 8, 16, 32, 64\}$ spanning six octaves, eight seeds each. We record whether training reaches zero error, the final norm, and the total number of weight updates (the physical learning-time observable; one epoch makes up to P updates, so update count, not epoch count, measures learning time).

Results (Fig. 11). *P1 holds:* every $U_{\text{th}} \geq 2$ reaches zero error in all eight seeds (the smallest, $U_{\text{th}} = 1$, in six of eight — consistent with a small-threshold edge effect rather than a failure of P1), confirming that existence is U_{th} -gauge across a $64\times$ range. *P2 holds, with structure:* the final norm is flat at a floor $\|w\| \approx 4.8$ for $U_{\text{th}} \leq 8$ and then grows linearly, $\|w\| \approx 0.39 U_{\text{th}}$, for $U_{\text{th}} \geq 16$ (linear-fit correlation $r = 0.995$). The floor is the natural magnitude of the small-norm solution the rule finds when the threshold does not force a larger one; only once U_{th} exceeds that scale does it drag the norm up in proportion. This floor-plus-linear structure is the dynamical counterpart of the small-norm/large-norm distinction that, under output noise, becomes the unbalanced/balanced dichotomy of [3] — a connection we note here and test directly only once noise is added (§11). *P3 fails as stated:* the naive law predicts an update count rising linearly with U_{th} (from $U_{\text{th}}/\lambda = 10$ at $U_{\text{th}} = 1$ to 640 at $U_{\text{th}} = 64$), but the measured count is instead nearly flat at ~ 140 updates for $U_{\text{th}} \leq 32$ — a range over which U_{th}/λ would have grown 32 -fold — and rises only to ~ 260 at $U_{\text{th}} = 64$. The learning time is dominated by a U_{th} -independent *direction-finding* cost set by the load; magnitude growth contributes only a sub-dominant term that becomes visible at the largest thresholds.

Discussion. The gauge invariance of existence (P1) and the norm scaling (P2) are confirmed; the naive learning-time law (P3) is refuted, and instructively so. The data resolve $\|w\|$ into a load-set floor plus a threshold-forced linear term — exactly the small-norm/large-norm distinction that

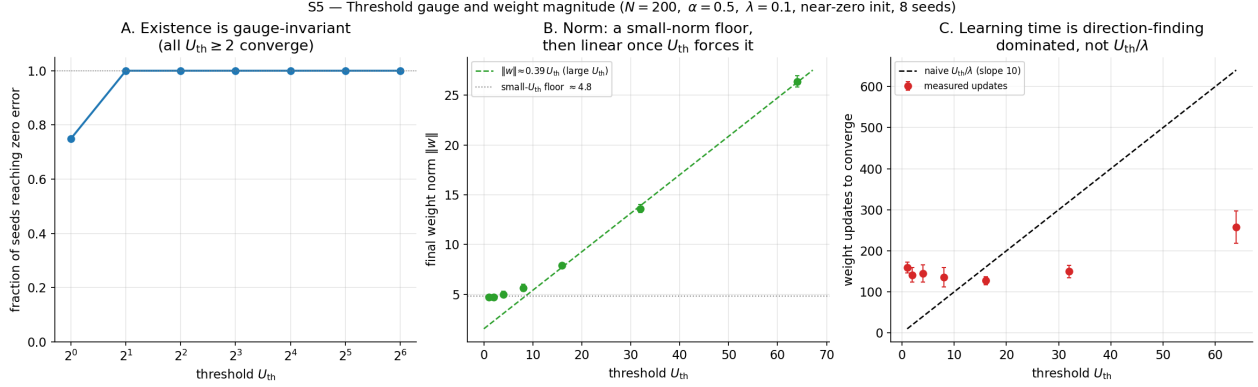


Figure 11: Threshold gauge and weight magnitude ($N = 200$, $\alpha = 0.5$, $\lambda = 0.1$, near-zero init, eight seeds). **(A)** Fraction of seeds reaching zero error vs U_{th} : existence is gauge-invariant ($U_{th} \geq 2$ all converge). **(B)** Final $\|w\|$ vs U_{th} : a small-norm floor (≈ 4.8), then linear growth ($\approx 0.39 U_{th}$) once the threshold dominates. **(C)** Weight updates to converge vs U_{th} : flat (direction-finding dominated), far below the naive U_{th}/λ line.

becomes the balanced/unbalanced dichotomy under noise [3]. That the learning time is direction-dominated rather than magnitude-dominated says the hard part of tempotron learning is locating the separating *direction* on the shattered surface, not scaling the weights to threshold — which is precisely the findability question (§11). A natural next step is to add output noise and look for the predicted collapse of the norm floor onto an $O(1)$ balanced solution, and to relate the direction-finding cost to the local geometry of the solution cluster the rule enters.

11 Discussion and open directions

What is established. The simulation pipeline is component-verified and the end-to-end measurement procedure is calibrated against three known answers (§7): the perceptron capacity $\alpha_c = 2$ and the Gardner margin, the Gumbel statistics of the peak voltage, and the kernel closed form. The threshold study (§10) confirms the gauge invariance of existence and resolves the weight norm into a load-set floor and a threshold-forced linear term, while showing that learning time is dominated by direction-finding rather than magnitude growth.

The findability question. For the perceptron, convexity and the perceptron-convergence theorem guarantee that a learning rule finds a separating w whenever one exists. For the tempotron the target (+1) constraints are non-convex (§5.1), so no such theorem applies, yet the bare rule reliably reaches zero error below capacity. Baldassi et al. [4] offer the mechanism for perceptron-like learners: simple algorithms are drawn not to the exponentially dominant *isolated* solutions but to rare, dense, high-local-entropy clusters. The direction-dominated learning time we measure (§10) is consistent with the rule’s cost being set by locating such a cluster; the magnitude/structure it settles on should then fingerprint which cluster it entered — the central, measurable target of this program.

Why extreme-value theory, not a central limit theorem. The decision variable is a maximum over time, not a sum, so its fluctuations are governed by the Fisher–Tippett–Gnedenko

theorem rather than the CLT; for a correlated Gaussian process the relevant law is Gumbel [9], which is the origin of the $\sqrt{2 \ln K}$ threshold scale and the slow $\ln \ln K$ growth of the capacity (§5.3).

A tractable exact $V_{\max}$. The grid scan of §5 is $O(|t|)$ per pattern. Between consecutive spikes only earlier spikes contribute, so $V(t) = A e^{-t/\tau_m} - B e^{-t/\tau_s}$ with $A, B = V_0 \sum_{t_i^f \leq t} w_i e^{t_i^f/\tau}$, whose interior stationary point is

$$t^* = \frac{\ln(A\tau_s/B\tau_m)}{1/\tau_m - 1/\tau_s}; \quad (20)$$

$V_{\max}$ is then the best over in-interval stationary points and spike onsets, an exact $O(\text{spikes log spikes})$ method (validated against the grid scan by reference-bisection) that makes the capacity sweep tractable.

Open directions. (i) Reproduce $\alpha_c(K)$ (RMS Fig. 2a) using exact linear feasibility for the half-crossing — whose cost, unlike an iterative max-margin solver, does not diverge near capacity (§7.1) — and the iterative oracle for the margin away from capacity; expect the slow $\sqrt{\ln K}$ approach with an additive offset α_0 . (ii) Settle whether the effective sample count is $T/\sqrt{\tau_s \tau_m}$ or T/τ_s (§7.2). (iii) Map the basin structure of the learning rule as α crosses α_c (connected $\rightarrow$ shattered) and test whether the rule prefers dense, robust clusters. (iv) Add output noise and look for the balanced $\leftrightarrow$ unbalanced norm transition (§10; [3]), measuring the robust capacity $\alpha_b \leq \alpha_c$.

12 Reproducibility and code map

Every number and figure regenerates from a seeded script under `experiments/tempotron_capacity/`. The model lives in `src/tempotron/` (`kernel.py`, `patterns.py`, `decision.py`, `learning.py`, `capacity.py`); the component verifications (§3–6) and the analytic references for the validation studies are gated by the `tests/` suite (`kernel`, `patterns`, `decision`, `learning`, `capacity`, and the separability oracle), which is `ruff`- and `strict-mypy`-clean; run `uv run pytest`. The validation and threshold studies (§7–10) run as seeded, config-determined batch jobs (`numpy/scipy` for the exact linear-feasibility oracle; `PyTorch` for the batched iterative oracle and the tempotron training), each writing a manifest that fixes the commit, environment, and random seed. Compute is modest: the perceptron sweep and the sanity/threshold studies each complete in minutes on a single consumer-class GPU, and the exact-oracle perceptron sweep runs on CPU. Figures rebuild with `uv run python figures/<name>.py`. Per-section theory notes are in `docs/` (01–03b, 05, and `sanity/`); the underlying capacity theory is consolidated in `docs/05-statistical-mechanics-of-capacity.md`.

References

- [1] R. Gütig and H. Sompolinsky, *The tempotron: a neuron that learns spike timing–based decisions*, Nat. Neurosci. **9**, 420–428 (2006).
- [2] R. Rubin, R. Monasson and H. Sompolinsky, *Theory of spike timing–based neural classifiers*, Phys. Rev. Lett. **105**, 218102 (2010); arXiv:1010.5496.
- [3] R. Rubin, L. F. Abbott and H. Sompolinsky, *Balanced excitation and inhibition are required for high-capacity, noise-robust neuronal selectivity*, Proc. Natl. Acad. Sci. **114**, E9366–E9375 (2017).

- [4] C. Baldassi, C. Borgs, J. Chayes, A. Ingrosso, C. Lucibello, L. Saglietti and R. Zecchina, *Unreasonable effectiveness of learning neural networks: From accessible states and robust ensembles to basic algorithmic schemes*, Proc. Natl. Acad. Sci. **113**, E7655–E7662 (2016); arXiv:1605.06444.
- [5] T. Zhang et al., *Origin of the efficiency of spike timing-based neural computation for processing temporal information*, Neural Networks (2022).
- [6] R. Gütiğ, *Spiking neurons can discover predictive features by aggregate-label learning*, Science **351**, aab4113 (2016).
- [7] T. M. Cover, *Geometrical and statistical properties of systems of linear inequalities...*, IEEE Trans. Electron. Comput. **EC-14**, 326–334 (1965).
- [8] E. Gardner, *The space of interactions in neural network models*, J. Phys. A **21**, 257 (1988).
- [9] M. R. Leadbetter, G. Lindgren and H. Rootzén, *Extremes and Related Properties of Random Sequences and Processes*, Springer (1983).
- [10] W. Krauth and M. Mézard, *Learning algorithms with optimal stability in neural networks*, J. Phys. A **20**, L745–L752 (1987).
- [11] W. Gerstner and W. M. Kistler, *Spiking Neuron Models*, Cambridge UP (2002).
- [12] Y. Nesterov, *Introductory Lectures on Convex Optimization: A Basic Course*, Springer (2004).
- [13] S. Bubeck, *Convex Optimization: Algorithms and Complexity*, Found. Trends Mach. Learn. **8**(3-4), 231–357 (2015).
- [14] J. K. Anlauf and M. Biehl, *The AdaTron — an adaptive perceptron algorithm*, Europhys. Lett. **10**(7), 687–692 (1989).
- [15] A. B. J. Novikoff, *On convergence proofs on perceptrons*, Symp. Math. Theory of Automata XII, 615–622 (1962).
- [16] M. Bouten, L. Reimers and B. Van Rompaey, *Capacity of perceptrons: finite-size corrections*, Phys. Rev. E **58**, 2386–2394 (1998).
- [17] A. Engel and C. Van den Broeck, *Statistical Mechanics of Learning*, Cambridge UP (2001).
- [18] E. Gardner and B. Derrida, *Optimal storage properties of neural network models*, J. Phys. A **21**, 271–284 (1988).
- [19] T. L. H. Watkin, A. Rau and M. Biehl, *The statistical mechanics of learning a rule*, Rev. Mod. Phys. **65**, 499–556 (1993).
- [20] J. Pickands III, *Asymptotic properties of the maximum in a stationary Gaussian process*, Trans. Amer. Math. Soc. **145**, 75–86 (1969).